Autonomous Poultry Farming in the UK: A Review of Technologies and Challenges

Nour Morsi
Robotics and Autonomous Systems Group, Department of Design, Manufacturing and Engineering Management
University of Strathclyde
Glasgow G1 1XJ, UK
nour.morsi@strath.ac.uk

Mohamed Khedr
Robotics and Autonomous Systems Group, Department of Design, Manufacturing and Engineering Management
University of Strathclyde
Glasgow G1 1XJ, UK
mohamed.khedr@strath.ac.uk

Erfu Yang
Robotics and Autonomous Systems Group, Department of Design, Manufacturing and Engineering Management
University of Strathclyde
Glasgow G1 1XJ, UK
erfu.yang@strath.ac.uk

Tariq Masood
Department of Design, Manufacturing and Engineering Management
University of Strathclyde
Glasgow G1 1XJ, UK
tariq.masood@strath.ac.uk

Anja Maier
Department of Design, Manufacturing and Engineering Management
University of Strathclyde
Glasgow G1 1XJ, UK
anja.maier@strath.ac.uk

Karen Campbell
Glenrath Farms
Edinburgh, EH45 8FF
karenc@glenrathfarms.co.uk

Ashley Wallace
Glenrath Farms
Edinburgh, EH45 8FF
ashley@glenrathfarms.co.uk

***Abstract*— As population growth accelerates, the global demand for eggs and poultry meat is anticipated to surge. However, the poultry industry encounters several challenges, including avian influenza outbreaks, economic inflation, and a diminishing skilled labour pool. These factors exert pressure on farmers to sustain efficiency and sustainability. Accordingly, the poultry industry in the United Kingdom is adopting advanced technologies such as computer vision, artificial intelligence (AI), and robotics to optimise operational processes. This paper presents an overview of the United Kingdom poultry industry, highlighting cutting-edge technologies, dominant challenges, and emerging trends. While smart technologies have demonstrated effectiveness in addressing certain challenges within the UK poultry industry, the adoption of more adaptable robotic systems and the implementation of AI-driven health monitoring systems are essential to attain the objective of establishing intelligent packing centres.**

Keywords*— *industry 4.0, robotics, poultry, artificial intelligence, gripper.

I. Introduction

With the projected global population reaching 9.2 billion by 2050, a scarcity of high-quality food is anticipated [1]. Therefore, the production of common foods such as eggs and poultry meat must be boosted to meet the escalating demand and achieve adequate food security levels. This can be achieved by optimising the supply chain to an enhanced and safer farm-to-plate process. The objective of this paper is to investigate the fundamental components of the egg and poultry industry, with a specific focus on its drivers, challenges, available technologies, and emerging trends shaping its future.

This review concentrates on the United Kingdom (UK) poultry industry, which adheres to stricter standards. For instance, the UK imposes the lowest stocking density (39 kg/m²) as compared to the European Union (EU) (42 kg/m²) [2], and the United States (US) (44 kg/m²) [3], which can be applied under certain conditions. Additionally, the United Kingdom banned the chlorine-washed chicken, which can conceal inadequate hygiene practices (as opposed to the US) [4].

UK egg production for human consumption has steadily increased in recent years. In Q1 2023, it reached 244 million dozen, and in Q1 2024, it rose to 248 million dozen, a 1.5% increase [5]. Production continued to grow in Q1 2025, reaching 255 million dozen, a 2.7% increase. UK egg consumption also increased, from 12.8 billion eggs in 2023 to 13.6 billion in 2024, indicating an increase in demand [6]. Although UK egg production is increasing, the country has yet to achieve full self-sufficiency, producing 87.3% of its consumption needs in 2023 and 88.2% in 2024, indicating progress but still some reliance on imports.

Consumers' increased demand for animal welfare and sustainability prompted farmers to prioritise producing healthy, high-quality animal products [7]. The UK government supported transitions such as abandoning cage-housing systems and extending the poultry production cycle [8], which aims to be completed by 2025 [9]. Despite these efforts, farmers face significant financial and logistical challenges, with 4 million laying hens still kept in cages [10].

To address this challenge, the implementation of digitisation and automation throughout every stage of the egg industry production cycle is imperative, from collection to delivery. The technologies introduced by the current industrial revolution (industry 4.0) can be the key to realising the future of smart packing centres. Industry 4.0 introduces higher levels of information integration and multi-industrial automation [1]. It includes technologies like artificial intelligence, digital twins, machine learning, and robotics. Industry 4.0 introduces intelligent machine communications to perform informed decisions with limited human involvement. A digitised production line offers increased productivity, resource efficiency and waste reduction [11].

This research is supported by Glenrath Farms and UK Research and Innovation (UKRI) under the Knowledge Transfer Partnership (KTP) project titled 'Development of an Intelligent Collaborative Robot System for Smart Farm Manufacturing (iCoBOTS)', project number KTP 13838.

The review conducted an exhaustive search of academic databases, including IEEE, Scopus, Web of Science, ScienceDirect, and Google Scholar, to identify scholarly publications on the poultry industry and egg handling that were published after 2020. In addition to academic research, commercially available technologies and prominent media outlets, such as Poultry News, were also scrutinised to gain comprehensive insights into the latest industry regulations and legislation. The search used keywords such as poultry industry, robotics, machine vision, machine learning, grippers, eggs, and food handling. The search remained broad enough to cover a range of technologies and applications within the poultry industry. The research questions are: (1) What are the key tasks in the egg & poultry industry and the challenges faced? (2) what are the technologies available to aid the egg & poultry industry to mitigate such challenges? (3) What are the future trends and challenges faced?

II. Egg and Poultry Production Overview

For profitable farming, a laying hen house must be maintained according to health and safety standards [12]. This includes (but not limited to) feeding, hen monitoring, cleaning, dead hen disposal, production packing, and management. These tasks, as shown in Fig.1, require intensive manual labour and significant time investment. Improper execution can lead to injuries and financial losses. Farm workers are also often exposed to dust and unpleasant odours, mainly from poultry manure, which releases organic gases such as ammonia and hydrogen sulphide [13]. Therefore, hen waste needs to be removed immediately to prevent the risk of disease among the hens.

Another critical factor affecting profitability is the cracking of eggs during packing and dispatch. Even though they might not be completely broken, micro-cracks can still develop; this can lead to bacterial invasion, rendering eggs unsuitable for human consumption [14]. The eggshell is a bio-ceramic composite structure, primarily made of limestone in the form of calcite. Due to the nature of the material, the shell is inherently brittle and prone to developing micro-cracks, particularly from impact with cage floors or mishandling. Its permeable shell allows for the exchange of respiratory gases and vapours while preventing microbial entry. Despite the shell thickness ranging between 350–450 µm, it can withstand forces of 30–50 N [15]. Estimates suggest that 8 to 11% of egg production losses are due to cracked eggs [16]. In the United States alone, these losses amount to over $247 million annually, emphasising that egg cracking remains a significant and costly issue in the poultry industry [15].

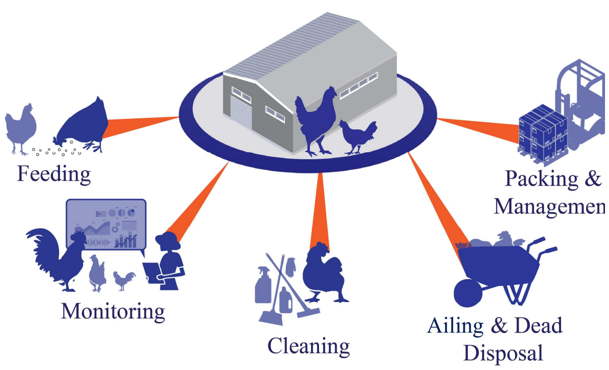

Fig. 1. Main processes in chicken farming house

TABLE I. SWOT Analysis of the Poultry Industry in the UK

Comparison	*Description*
Strengths	Egg consumption is rising. Technology ensures consistent production rates. Robotics helps address skilled labour shortages. Robotics enables worker upskilling and productivity gains. Robotics helps maintain hygiene standards. Robotics reduces production defects.
Weaknesses	Switching from cages to barns is costly. Technology increases electrical costs. Robotic systems have limited flexibility. Cybersecurity risks (e.g. hacking) exist. Skilled workers are hard to recruit. Standards vary across companies and foreign staff. Training workers on robotics is difficult.
Opportunities	Sensor data supports better decision-making. Consumer education improves perception of technology.
Threats	Evolving trade policies increase political pressure. Staff reluctance to adopt robotic systems. Safety limitations in dynamic environments. System breakdowns cause order delays. Climate change increases poultry mortality. Ukraine war raises production costs. Risk of cross-contamination from robotic grippers. Avian flu remains a major production threat.

III. Current Limitations

The UK poultry industry is currently facing several compounding challenges, a SWOT analysis of the poultry industry in the UK based on current events is shown in Table I. From the analysis the challenges are: (1) A reduction in skilled labour following Brexit (2) A decline in colony egg production (3) Escalating input costs due to world events (such as the war in Ukraine) (4) Inflationary pressures (5) A record outbreak of avian influenza.

One of the primary issues affecting production is the shortage of staff. This problem has been widely publicised in national news outlets [17]. By 2023, the top 20 egg producers in the UK reported a combined turnover of £1 billion, representing a 1.4% increase from the previous year [18]. Despite this, profitability remains under pressure. For example, in 2021, Noble Foods, the market leader, reported losses of £32.6 million, a deeper loss than the £30 million deficit the year before [18]. In contrast, Glenrath Farms posted a modest profit of £4.1 million. In 2023, Noble foods reported a turnover of a £6.3 million loss [19]. On the other hand, Glenrath farms reported an increased profit £6.9 million [20].

Additionally, the avian influenza outbreak of 2022 had a significant impact on flock numbers. The UK's laying hen population declined from 43 million in 2021 to 38 million in 2022 [21]. However, by June 2024, the number of laying hens had slightly recovered to about 40 million [21]. External factors such as the cost-of-living crisis, and the war in Ukraine have exacerbated the situation. Currently, the US influenza outbreak has wiped 30 million US laying hens [22], leading to the US seeking imports from European countries such as Poland and Turkey. Such impact will indirectly affect the global economy, including the UK.

In 2022, the Ukraine war contributed to a 40% increase in wheat feed costs, this has left farmers struggling [23], as wheat constitute a component of the laying hen feed, which accounts for 60% of the egg production costs [24]. Although egg prices eventually rose to offset this hike, the initial response was slow. Over time, prices increased by 85% compared to the previous year, but this lag meant producers were unable to recoup losses, which totalled £20.3 million. In response, some retailers, including Tesco, have offered production support to the British egg industry [25].

Looking ahead, the agricultural sector must find ways to mitigate systemic risks, particularly those introduced by climate change such as water scarcity [26]. Addressing these risks requires action at the farm level. Broader disruptions such as the COVID-19 pandemic, population growth, and geopolitical instability also threaten food security, underscoring the need for resilient, adaptive agri-food systems.

IV. Cutting Edge Technologies in Poultry Industry

A. *Computer Vision & Machine Learning*

Computer vision is a process of applying mathematical formulations to digital images with the aim to achieve what human visual inspection can do [27]. Specifically, machine vision facilitates automated inspection, monitoring, and guiding of machines such as robots. It has been a rich research topic in animal monitoring as they offer non-intrusive and non-invasive supervision techniques [28]. On top of that, the approach reduces laborious processes. The main components of a vision system include a vision camera and the processing algorithm. With all the advantages described, computer vision performance varies with light conditions, occlusions, etc.

Combined with machine vision, machine learning (ML) is a subcategory of artificial intelligence. ML systems can complete a task without being completely hard coded. This is done through training the system to learn the relationships using desired results. Using the desired results, the system changes its configuration to fit the required result [29].

Machine learning-based flexible systems can deal with semi-structured environments and environments that exhibit minimal changes. In effect, systems introduce reactions autonomously based on training datasets. Even though such an approach improves system flexibility, it can be susceptible to bias and overfitting issues, a condition in which the system has memorised the patterns instead of generalising from them. Therefore, the quality and balance of training datasets are crucial components to building a system. Also, datasets must be regularly updated to support emerging challenges such as new outbreaks and handling new pathogens [30]. Fig. 2 shows an overview of an autonomous poultry breeding system.

1) *Cracked Egg Detection.*

Several approaches to detect cracked eggs have been proposed. In [31], a Convolutional Neural Network (CNN)-based method was used to detect micro cracks using a patch-based approach with data augmentation. Firstly, images were enhanced using Contrast-Limited Adaptive Histogram Equalisation (CLAHE) and Sobel edge detection; a binary mask using Ostu's thresholding was used to isolate the egg surface. From the egg, the ten longest edges were extracted to be used as a dataset for the CNN model training. The images used to train the machine learning system were split into two classes: crack (all types of cracks) and intact (non-crack patches, calcium deposits, and stripes). The system achieved a 95.38% whole egg accuracy, outperforming SVMs using traditional features.

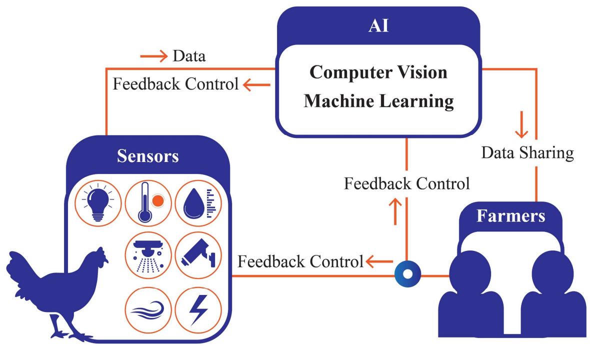

Fig. 2. Schematic diagram of perception and control in modern unmanned poultry breeding

In [32], a combination of supervised machine learning and deep learning was used for egg classification and weighting. A real-time multitask detection (RTMDet) and random forest model were develped to identify egg defects, weight, and category. RTMDet, a CNN-based architecture, was used to extract egg features such as the major and minor axes for classification into five categories (intact, crack, bloody, floor, and non-standard) while the random forest model predicted the egg weight based on the extracted features, achieving a minimum accuracy of 94% [32].

2) *Health & Disease Detection*

Common poultry diseases include infectious bronchitis, avian influenza, and infectious sinusitis. Checks for diseases include posture, feathers, faeces, and sound. In turn, inspector expertise is crucial for accurate checks. To mitigate the shortage of skilled labour, machine learning models have been proposed for disease detection.

In [33], poultry droppings were classified into six categories based on their visual abnormalities. To create a dataset, a pan-tilt zoom camera was installed on the farm in Lithuania, scanning for litter-covered surfaces. Images captured by the camera are transferred to an image processing model for segmentation (using the K-means algorithm) and classification using three different models (Resnet101, modified VGG-16, and YOLOv5). Fig. 3 shows the categories of the faeces selected based on the disease exhibited:

- Significant amount of moisture and discolouration: diarrhoea.
- High moisture and gas: digestive disturbance.
- Undigested feed: signs of malabsorption.

YOLOv5 approach has achieved the best performance, with a 91% classification accuracy [33]. It is important to note that a small amount of data was collected (due to the intensity of the task); as such, data augmentation techniques were performed to create enough data to train the model. Data augmentation focuses on increasing the volume, quality, and diversity of the data available [34].

In [35], a dataset containing the potential spread and distribution of avian influenza cases in Ireland, compiled from 1980 to 2020, was used to develop a machine learning model for early detection of the disease. Data preprocessing included the removal of outliers, null values, and duplicate entries, along with label encoding of categorical features. The cleaned dataset was then used to train several machine learning models, including Decision Tree, Light Gradient Boosting Machine (LGBM), Random Forest, CATBoost, and others. Based on accuracy metrics, the Decision Tree and LGBM models performed the best, achieving an accuracy of

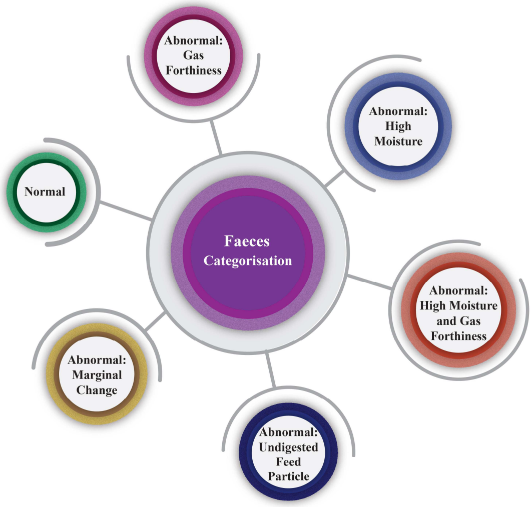

Fig. 3. Faeces categorisation

99.66%. The study's approach demonstrates the potential of machine learning as a valuable tool for rapid outbreak response and effective control of avian influenza [34].

B. Robotics

The rapid adoption of robotics in the poultry industry has been significantly accelerated by the integration of machine vision and artificial intelligence techniques. In [36], a disinfecting mobile robot was developed as shown in Fig. 4. The system comprised a mobile vehicle, spraying equipment, sensors, and a controller. The system utilised magnet-RFID navigation for reliable path tracking. Using a custom-designed air nozzle, the system provides a smart automated disinfection system. The system's motion and performance were tested in a poultry house. The robot was able to automatically navigate (at speeds 0.1-0.5 m/s) spraying liquid disinfectant at a rate of 200-400 mL/min.

In [37], an autonomous inspection robot was developed as an approach to reduce human labour and enhance hygiene. The robot detects and removes deceased hens in caged layer houses. It uses QR code-based localization, autonomous navigation via magnetic strips, and an RTMDet model for real-time hen detection. With a detection speed of 9 meters per minute, the system achieved 90.61% accuracy and a 0.14% false detection rate. While the system performed with sufficient accuracy, complex farming environments and varying lighting conditions still affect its performance.

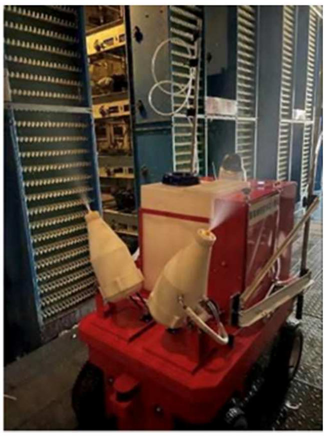

Fig. 4. Disinfecting robot proposed by [36]. ©2021 IEEE

In [38], the authors explored mobile robotic system design, featuring functions such as chicken behaviour recognition, as shown in Fig. 5, floor egg collection, and dead chicken removal. The introduced system integrates a YOLOv7 AI model for detection. According to the authors, the system offers a more adaptable approach compared to other models.

After getting the eggs palletised, eggs are sent to the packing station, they are graded by an egg grader such as the Omnia PX [39]. Egg graders are large sorting machines that classify the eggs into their weight classes and remove damaged or defective eggs. They are then assigned to lanes where they are packed in carton boxes for retailers. However, such machines still rely on manual labour. Approaches to mitigate this include having robotic systems connected at the end of the line for packing such as the MR12 case packer [40].

1) *Grippers*

An important component of robotic systems to pick up eggs are robotic grippers. Robotic grippers constitute critical components within robotic systems, serving as the primary mechanism for enabling automation through physical interaction with objects across diverse application contexts. They have a remarkable variation in their shape, functionality, and material to meet the requirements of the intended application, with their versatility enabling them to address unique challenges across numerous industries, particularly in terms of adaptability and payload capacity. A well-designed gripper can significantly enhance the robot's capabilities, allowing it to perform complex tasks with precision and efficiency. Grippers in the food industry face the challenge of handling gentle, irregular, and sensitive items. Food safety must be ensured, minimal damage must be maintained, and hygiene standards must be adhered to while performing at high speeds.

In [41], an adaptable vacuum system with the KUKA KR 60-3 robot was introduced. It is configured in a 5x6 matrix of suction cups to act as a gripper for egg handling. The gripper packed 660 eggs across 40 tests without damage, achieving 100% efficiency in controlled tests, with a maximum payload of up to 30 eggs (around 6 kg), and the ability to limit the applied vacuum pressure to avoid damaging the eggshells. Despite its strengths, the experiments in [41] were conducted exclusively in a controlled laboratory environment, leaving the gripper's performance in dynamic or production-line conditions untested. Furthermore, energy efficiency is not optimised, and there is little discussion on the durability or material wear of the gripper components over time.

In [42], a three-finger soft robotic gripper was developed. It is powered using a stepper motor and a pulley mechanism to grasp delicate and irregular objects adaptively. Weighing 1.2 kg, the gripper can lift objects up to 4.2 kg and 9.5 kg with an anti-slip foam tape. As an experiment to demonstrate autonomous grasping, the gripper was mounted on a SCARA robot with an RGB-D camera and a NVIDIA Jetson TX2 used

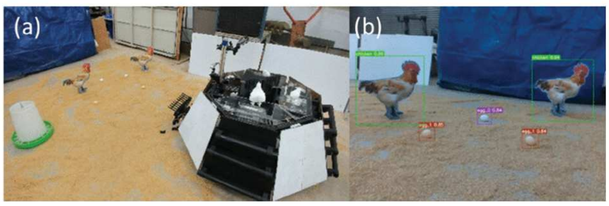

Fig. 5. (a) mobile robot design (b) object detection proposed by [38]. ©2024 IEEE

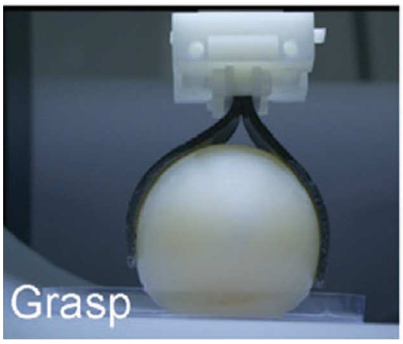

Fig. 6. Compliant gripper designed to pick up delicate objects by [43]. Used with permission of John Wiley and Sons from 'Porous Magnetic Soft Grippers for Fast and Gentle Grasping of Delicate Living Objects'. Authored by Mujun Li, Bing Hu, Shiwu Zhang, et al. permission conveyed through Copyright Clearance Center, Inc.

for image processing during grasping. The gripper successfully handled eggs, fruits, ceramics, and glass products, proving its suitability for tasks requiring gentle yet reliable manipulation.

In [43], a novel design approach by integrating a porous structure into a magnetic silicon elastomer was introduced, as shown in Fig. 6. Creating a gripper with enhanced compliance made for the fast and gentle grasping of delicate living objects. The authors addressed the limitations of increasing magnetic particle concentration to enhance magnetisation strength by introducing porous structures into the gripper to reduce the elastic modulus while maintaining the magnetisation strength. According to the authors, the gripper exhibits outstanding performance characteristics, including high softness (modulus as low as 0.062 MPa), remarkable adaptability, a high lifting ratio (30 times under 54 mT), and a rapid response. The gripper can catch a thrown raw quail egg in 70 milliseconds, which is faster and more precise than most grippers. However, the authors did not discuss the gripper's long-term durability and performance under repeated use. They also didn't consider whether the mechanical properties of the porous structure and the magnetic properties of the composite material may degrade over time, affecting the gripper's performance.

V. Challenges & Future Trends

While the integration of industry 4.0 technologies, such as robotics and machine learning, has progressed through single application-specific implementations, significant challenges persist broader, system level integration.

A. *Emerging Technologies in Soft and Adaptive Grippers*

Robotic grippers for egg handling are evolving towards softer, more adaptable, and intelligent designs. While vacuum, electro-adhesive, magnetic, and pneumatic systems each have advantages, they also have trade-offs in terms of durability, control complexity, hygiene, and scalability. It is important to note that grippers for picking up egg trays and packs are rarely explored due to pack design challenges. Pack locking mechanisms prevent the use of vacuum grippers, and the gripper mechanism must handle egg weight and contact at corners without breaking them. The design must also accommodate varying pack weights and sizes, as well as clearance during pick and place.

B. *Robotic Systems Adaptability*

Robotic systems introduced for poultry farming are still in their early development, with most robotic systems designed for specific, singular tasks such as egg collection or dead bird removal, with few capable of performing multiple functions. As a result, current solutions do not yet meet the industry's broader demands for efficiency, flexibility, and animal welfare. Other challenges include navigation in dynamic environments, safe interactions with animals, and weight limitations [38].

C. *AI Dependency on Human Preference*

Farmers require a comprehensive assessment of an animal's mental, biological, and functional states. However, current AI algorithms often focus on monitoring isolated aspects of animal health, falling short of providing a holistic view. Furthermore, the use of AI still relies heavily on human input to define the metrics used for health assessment. This includes training algorithms on either pre-existing or custom-made datasets, which can introduce inaccuracies such as biases during training, potentially compromising the reliability of the monitoring system [44].

VI. Conclusion

In summary, the UK poultry industry is pressured from understaffing, avian flu outbreaks, and feed costs. Smart technologies such as robotics, computer vision, and machine learning offer a way to tackle such challenges. However, such technologies still face deployment issues in real environments. This is due to the ever-changing environment (pathogens, regulatory changes, etc.) which requires up-to-date datasets (to train machine learning models) and robotic systems to be adaptable to product changes. Since technologies such as AI models rely on large datasets, enhanced collaboration between research institutions is essential. Future research should focus on building upon the current advances to develop multipurpose robotics, robust grippers, and AI systems less reliant on human intervention to combat avian flu outbreaks to achieve sustainable smart packing centres.

Acknowledgment

We would like to thank Serena Broadway (UK Innovate Knowledge Transfer Advisor) and Dino Bertolaccini (University of Strathclyde Technician) for their kind support.

References

[1] M. W. Ahmed, S. J. Hossainy, A. Khaliduzzaman, J. L. Emmert, and M. Kamruzzaman, "Non-destructive optical sensing technologies for advancing the egg industry toward Industry 4.0: A review," COMP REV FOOD SCI F, vol. 22, no. 6, pp. 4378-4403, 2023, doi: 10.1111/1541-4337.13227.

[2] CIWF, "Information sheet - Broiler chicken production in the EU and UK," Compassion in Food Business, Nov. 2023. [Online]. Available: https://www.compassioninfoodbusiness.com/media/7455892/info-sheet-1-broiler-production-eu-and-uk.pdf.

[3] NCC, "NCC Animal welfare guidelines and audit checklist for broilers," National Chicken Council, Jul. 2017. [Online]. Available: https://www.nationalchickencouncil.org/wp-content/uploads/2017/07/NCC-Welfare-Guidelines-Broilers.pdf.

[4] RSPCA, "What is chlorinated chicken?," RSPCA Assured. [Online]. Available: https://www.rspcaassured.org.uk/farmed-animal-welfare/chickens/what-is-chlorinated-chicken/.

[5] S. Marsh, and H. Horton, "Low-welfare eggs from caged hens imported to UK in "staggering" numbers," The Guardian, Oct. 2023. [Online]. Available: https://www.theguardian.com/business/2023/oct/24/low-welfare-eggs-from-caged-hens-imported-to-uk-in-staggering-numbers.

[6] British Egg Industry Council, "Egg facts and figures," *Egg Info*. [Online]. Available: https://www.egginfo.co.uk/egg-facts-and-figures/industry-information/data.

[7] J. Gautron, S. Réhault-Godbert, T. G. H. Van de Braak, and I. C. Dunn, “Review: What are the challenges facing the table egg industry in the next decades and what can be done to address them?,” ANIMAL, vol. 15, pp. 100282-100282, 2021, doi: 10.1016/j.animal.2021.100282.

[8] G. Scrinis, C. Parker, and R. Carey, “The caged chicken or the free-range egg? The regulatory and market dynamics of layer-hen welfare in the UK, Australia and the USA,” J Agric Environ Ethics, vol. 30, no. 6, pp. 783-808, 2017, doi: 10.1007/s10806-017-9699-y.

[9] T. Mcdougal, “Is the UK egg sector on track to be cage-free by 2025?,” Poultry World, Dec. 2024. [Online]. Available: https://www.poultryworld.net/poultry/layers/is-the-uk-on-track-to-be-cage-free-by-2025.

[10] D. Corredor, “Will the UK egg industry achieve cage-free status by 2025?,” aviNews, Dec. 2024. [Online]. Available: https://avinews.com/en/will-the-uk-egg-industry-achieve-cage-free-status-by-2025/.

[11] M. Ghobakhloo, “Industry 4.0, digitization, and opportunities for sustainability,” J CLEAN PROD, vol. 252, p. 119869, 2020, doi: 10.1016/j.jclepro.2019.119869.

[12] DEFRA, “Code of practice for the welfare of laying hens and pullets,” GOV.UK, Aug. 2018. [Online]. Available: https://assets.publishing.service.gov.uk/media/5b69652be5274a1518c298db/code-of-practice-welfare-of-laying-hens-pullets.pdf.

[13] D. D. Sanjaya and A. Fadlil, “Monitoring temperature and humidity of boiler chicken cages based on Internet of Things (IoT),” Buletin Ilmiah Sarjana Teknik Elektro, vol. 5, no. 2, pp. 180-189, 2023, doi: 10.12928/biste.v5i2.4897.

[14] G. Kulshreshtha et al., “Effect of egg washing and hen age on cuticle quality and bacterial adherence in table eggs,” MICROORGANISMS, vol. 12, no. 10, p. 2027, 2024, doi: 10.3390/microorganisms12102027.

[15] R. M. G. Hamilton and W. L. Bryden, “Relationship between egg shell breakage and laying hen housing systems - an overview,” WORLD POULTRY SCI J, vol. 77, no. 2, pp. 249-266, 2021, doi: 10.1080/00439339.2021.1878480.

[16] K. Wengerska, J. Batkowska, and K. Drabik, “The eggshell defect as a factor affecting the egg quality after storage,” Poultry Science, vol. 102, no. 7, Art. no. 102749, 2023, doi: 10.1016/j.psj.2023.102749.

[17] R. Shields, “Crisis and opportunity: How UK farming can overcome labour shortages and soaring costs,” Agricultural Recruitment Specialists, Apr. 2025. [Online]. Available: https://www.agrirs.co.uk/blog/2025/04/crisis-and-opportunity-how-uk-farming-can-overcome-labour-shortages-and-soaring-costs,

[18] G. Duncan, “Cracking up: profit/loss of top 10 egg packagers,” The Grocer, Feb. 2023. [Online]. Available: https://www.thegrocer.co.uk/reports-and-data/cracking-up-profit/loss-of-top-10-egg-packagers/676014.article.

[19] C. House, “Glenrath Farms Ltd.,” GOV.UK. [Online]. Available: https://find-and-update.company-information.service.gov.uk/company/SC042166/filing-history.

[20] C. House, “Noble Foods Limited,” GOV.UK. [Online]. Available: https://find-and-update.company-information.service.gov.uk/company/03636168.

[21] C. Reeve, “UK poultry flock rebounds in 2024 as egg volumes pick up,” Farmers Weekly, Jul. 2024. [Online]. Available: https://www.fwi.co.uk/livestock/poultry/uk-poultry-flock-rebounds-in-2024-as-egg-volumes-pick-up.

[22] C. McCormack, “Bird flu forced 30M chickens to be culled across nine states so far this year,” New York Post, Apr. 2025. [Online]. Available: https://nypost.com/2025/04/22/us-news/bird-flu-forced-30m-chickens-to-be-culled-across-nine-states-so-far-this-year/.

[23] C. Mclaughlin, “How spiralling cost of chicken feed is driving UK egg crisis: Boss of poultry feed firm says wheat shortage caused by Ukraine War and Europe’s dry weather is pushing up prices to ‘record’ levels leaving egg farmers unable to meet production costs,” Mail Online. May2022.[Online].Available: https://www.dailymail.co.uk/news/article-10808605/How-spiralling-cost-chicken-feed-driving-UK-egg-crisis-Massey-Feeds-explains.html.

[24] BFREPA, “Fairness in the food supply chain enquiry,” [Online]. Available: https://committees.parliament.uk/writtenevidence/122895/pdf.

[25] Tesco, “Tesco announces close to £14 million of additional support for British egg industry,” Tesco, Nov. 2022. [Online]. Available: https://www.tescoplc.com/tesco-announces-close-to-14-million-of-additional-support-for-british-egg-industry/.

[26] Y. A. Attia et al., “Climate change and its effects on poultry industry and sustainability,” Discov Sustain, vol. 5, no. 1, pp. 397-22, 2024, doi: 10.1007/s43621-024-00627-2.

[27] N. S. N. Abd Aziz, S. Mohd Daud, R. A. Dziyauddin, M. Z. Adam, and A. Azizan, “A review on computer vision technology for monitoring poultry farm-application, hardware, and software,” Access, vol. 9, pp. 12431-12445, 2021, doi: 10.1109/ACCESS.2020.3047818.

[28] C. Okinda et al., “A review on computer vision systems in monitoring of poultry: A welfare perspective,” ARTIF INTELL AGR, vol. 4, pp. 184-208, 2020, doi: 10.1016/j.aiia.2020.09.002.

[29] I. El Naqa and M. J. Murphy, “What is machine learning?” pp. 3-11, 2015, doi: 10.1007/978-3-319-18305-3_1.

[30] R. O. Ojo, A. O. Ajayi, H. A. Owolabi, L. O. Oyedele, and L. A. Akanbi, “Internet of things and machine learning techniques in poultry health and welfare management: A systematic literature review,” Computers and Electronics in Agriculture, vol. 200, p. 107266, 2022, doi: 10.1016/j.compag.2022.107266.

[31] B. Botta, S. S. R. Gattam, and A. K. Datta, “Eggshell crack detection using deep convolutional neural networks,” Journal of Food Engineering, vol. 315, Art. no. 110798, 2022, doi: 10.1016/j.jfoodeng.2021.110798.

[32] X. Yang, R. B. Bist, S. Subedi, and L. Chai, “A computer vision-based automatic system for egg grading and defect detection,” ANIMALS-BASEL, vol. 13, no. 14, p. 2354, 2023, doi: 10.3390/ani13142354.

[33] A. Nakrosis et al., “Towards early poultry health prediction through non-invasive and computer vision-based dropping classification,” ANIMALS-BASEL, vol. 13, no. 19, p. 3041, 2023, doi: 10.3390/ani13193041.

[34] A. Mumuni and F. Mumuni, “Data augmentation: A comprehensive survey of modern approaches,” Array, vol. 16, p. 100258, 2022, doi: 10.1016/j.array.2022.100258.

[35] T. Choudhury, D. Ghosh, M. K. Gourisaria, J. J. Jena, P. Pattnayak, and A. Bandyopadhyay, “Revolutionizing H5_HPAI Detection: Role of Machine Learning in Early Diagnosis,” 2024 Second International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI), Coimbatore, India, 2024, pp. 882–887, doi: 10.1109/ICoICI62503.2024.10696247.

[36] Q. Feng, B. Wang, W. Zhang, and X. Li, “Development and test of spraying robot for anti-epidemic and disinfection in animal housing,” WRC SARA, pp. 24-29, 2021, doi: 10.1109/WRCSARA53879.2021.9612617.

[37] W. Ma et al., “Autonomous inspection robot for dead laying hens in caged layer house,” Computers and Electronics in Agriculture, vol. 227, p. 109595, 2024, doi: 10.1016/j.compag.2024.109595.

[38] C. L. Chang, R. Y. Xu, and J. Y. Ke, “Design and implementation of an AI-driven mobile robot for managing poultry tasks,” CACS, pp. 1-6, 2024, doi: 10.1109/CACS63404.2024.10773178.

[39] M. Group, “Omnia PX,” Moba, [Online]. Available: https://www.moba.net/page/en/Products/Detail/omnia-px/5.

[40] M. Group, “MR12 case packer,” Moba, [Online]. Available: https://www.moba.net/page/en/Products/Detail/mr12-case-packer/387.

[41] B. Carderon, L. Vinces, and J. Oliden, “An adaptable vacuum gripper design for egg packaging, attached to the KUKA KR 60-3 industrial robot,” LACCEI, vol. 1, no. 8, 2023, doi: 10.18687/LACCEI2023.1.1.834.

[42] C. -H. Liu, F. -M. Chung, Y. Chen, C. -H. Chiu and T. -L. Chen, “Optimal Design of a Motor-Driven Three-Finger Soft Robotic Gripper,” in IEEE/ASME Transactions on Mechatronics, vol. 25, no. 4, pp. 1830-1840, Aug. 2020, doi: 10.1109/TMECH.2020.2997743.

[43] X. Li et al., “Porous magnetic soft grippers for fast and gentle grasping of delicate living objects,” ADV MATER, vol. 36, no. 44, pp. e2409173-n/a, 2024, doi: 10.1002/adma.202409173.

[44] H. He, J. Gray, A. Cangelosi, Q. Meng, T. M. McGinnity and J. Mehnen, “The Challenges and Opportunities of Human-Centered AI for Trustworthy Robots and Autonomous Systems,” in IEEE Transactions on Cognitive and Developmental Systems, vol. 14, no. 4, pp. 1398-1412, Dec. 2022, doi: 10.1109/TCDS.2021.3132282.